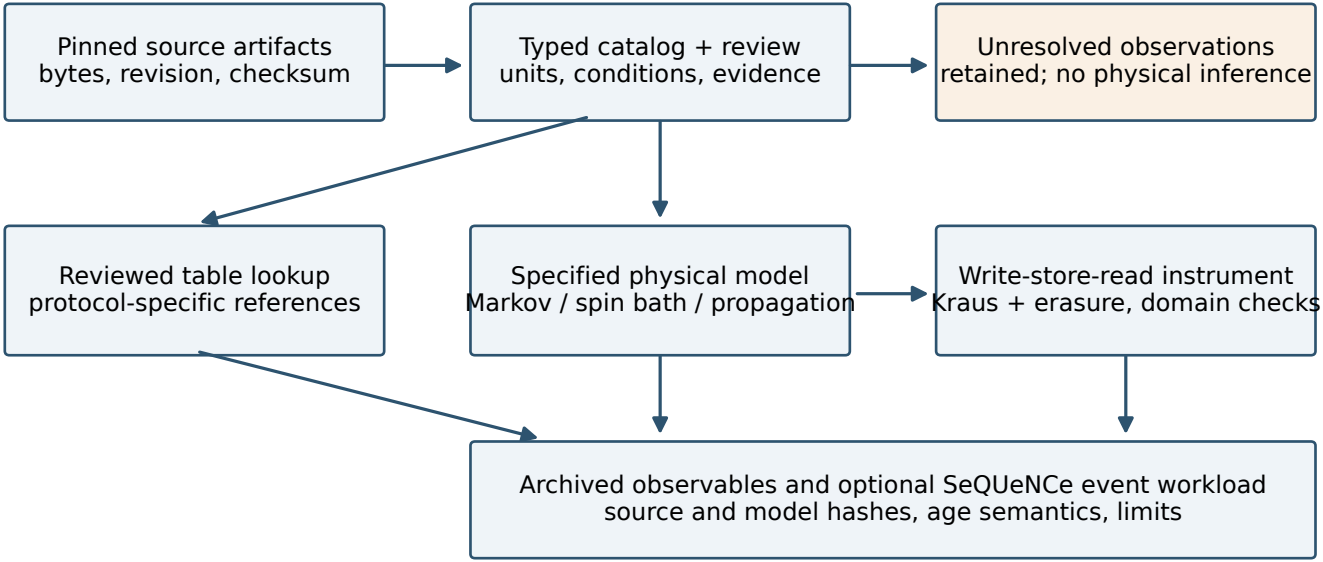
Pinned source artifacts
bytes, revision, checksum
Typed catalog + review
units, conditions, evidence
Unresolved observations
retained; no physical inference
Reviewed table lookup
protocol-specific references
Specified physical model
Markov / spin bath / propagation
Write-store-read instrument
Kraus + erasure, domain checks
Archived observables and optional SeQUeNCe event workload
source and model hashes, age semantics, limits